

9.1.7 Input configuration

When the I/O Port is programmed as Input:

- ☐ The Output Buffer is disabled
- ☐ The Schmitt Trigger Input is activated
- ☐ The weak pull-up and pull-down resistors are activated or not depending on input configuration (pull-up, pull-down or floating):
- ☐ The data present on the I/O pin is sampled into the Input Data register every APB2 clock cycle
- ☐ A read access to the Input Data register obtains the I/O

9.1.8 Output configuration

When the I/O Port is programmed as Output:

- ☐ The Output Buffer is enabled:
 - Open Drain Mode: A “0” in the Output register activates the N-MOS while a “1” in the Output register leaves the port in Hi-Z (the P-MOS is never activated)
 - Push-Pull Mode: A “0” in the Output register activates the N-MOS while a “1” in the Output register activates the P-MOS
- ☐ The Schmitt Trigger Input is activated.
- ☐ The weak pull-up and pull-down resistors are disabled.
- ☐ The data present on the I/O pin is sampled into the Input Data register every APB2 clock cycle
- ☐ A read access to the Input Data register gets the I/O state in open drain mode
- ☐ A read access to the Output Data register gets the last written value in Push-Pull mode

9.2 GPIO registers

Refer to *Section 2.2 on page 45* for a list of abbreviations used in register descriptions.

The peripheral registers have to be accessed by words (32-bit).

9.2.1 Port configuration register low (GPIOx_CRL) (x=A..G)

Address offset: 0x00

Reset value: 0x4444 4444

Bits 31:30, 27:26,

23:22, 19:18, 15:14,

11:10, 7:6, 3:2

CNFy[1:0]: Port x configuration bits (y= 0 .. 7)

These bits are written by software to configure the corresponding I/O port.

Refer to *Table 20: Port bit configuration table*.

In input mode (MODE[1:0]=00):

00: Analog mode

01: Floating input (reset state)

10: Input with pull-up / pull-down

11: Reserved

In output mode (MODE[1:0] > 00):

00: General purpose output push-pull

01: General purpose output Open-drain

10: Alternate function output Push-pull

11: Alternate function output Open-drain

Bits 29:28, 25:24,

21:20, 17:16, 13:12,

9:8, 5:4, 1:0

MODEy[1:0]: Port x mode bits (y= 0 .. 7)

These bits are written by software to configure the corresponding I/O port.

Refer to *Table 20: Port bit configuration table*.

00: Input mode (reset state)

01: Output mode, max speed 10 MHz.

10: Output mode, max speed 2 MHz.

11: Output mode, max speed 50 MHz.

9.2.2 Port configuration register high (GPIOx_CRH) (x=A..G)

Address offset: 0x04

Reset value: 0x4444 4444

Bits 31:30, 27:26,

23:22, 19:18, 15:14,

11:10, 7:6, 3:2

CNFy[1:0]: Port x configuration bits (y= 8 .. 15)

These bits are written by software to configure the corresponding I/O port.

Refer to *Table 20: Port bit configuration table*.

In input mode (MODE[1:0]=00):

00: Analog mode

01: Floating input (reset state)

10: Input with pull-up / pull-down

11: Reserved

In output mode (MODE[1:0] > 00):

00: General purpose output push-pull

01: General purpose output Open-drain

10: Alternate function output Push-pull

11: Alternate function output Open-drain

Bits 29:28, 25:24,

21:20, 17:16, 13:12,

9:8, 5:4, 1:0

MODEy[1:0]: Port x mode bits (y= 8 .. 15)

These bits are written by software to configure the corresponding I/O port.

Refer to *Table 20: Port bit configuration table*.

00: Input mode (reset state)

01: Output mode, max speed 10 MHz.

10: Output mode, max speed 2 MHz.

11: Output mode, max speed 50 MHz.

9.2.3 Port input data register (GPIOx_IDR) (x=A..G)

Address offset: 0x08h

Reset value: 0x0000 XXXX

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **IDRy**: Port input data (y= 0 .. 15)

These bits are read only and can be accessed in Word mode only. They contain the input value of the corresponding I/O port.

9.2.4 Port output data register (GPIOx_ODR) (x=A..G)

Address offset: 0x0C

Reset value: 0x0000 0000

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **ODRy**: Port output data (y= 0 .. 15)

These bits can be read and written by software and can be accessed in Word mode only.

Note: For atomic bit set/reset, the ODR bits can be individually set and cleared by writing to

the GPIOx_BSRR register (x = A .. G).

9.2.5 Port bit set/reset register (GPIOx_BSRR) (x=A..G)

Address offset: 0x10

Reset value: 0x0000 0000

Bits 31:16 **BRy**: Port x Reset *bit* y (y= 0 .. 15)

These bits are write-only and can be accessed in Word mode only.

0: No action on the corresponding ODRx bit

1: Reset the corresponding ODRx bit

Note: If both BSx and BRx are set, BSx has priority.

Bits 15:0 **BSy**: Port x Set bit y (y= 0 .. 15)

These bits are write-only and can be accessed in Word mode only.

0: No action on the corresponding ODRx bit

1: Set the corresponding ODRx bit

9.2.6 Port bit reset register (GPIOx_BRR) (x=A..G)

Address offset: 0x14

Reset value: 0x0000 0000

Bits 31:16 Reserved

Bits 15:0 **BRy**: Port x Reset bit y (y= 0 .. 15)

These bits are write-only and can be accessed in Word mode only.

0: No action on the corresponding ODRx bit

1: Reset the corresponding ODRx bit

9.2.7 Port configuration lock register (GPIOx_LCKR) (x=A..G)

This register is used to lock the configuration of the port bits when a correct write sequence

is applied to bit 16 (LCKK). The value of bits [15:0] is used to lock the configuration of the GPIO. During the write sequence, the value of LCKR[15:0] must not change. When the LOCK sequence has been applied on a port bit it is no longer possible to modify the value of

the port bit until the next reset.

Each lock bit freezes the corresponding 4 bits of the control register (CRL, CRH).

Address offset: 0x18

Reset value: 0x0000 0000

Bits 31:17 Reserved

Bit 16 **LCKK[16]**: Lock key

This bit can be read anytime. It can only be modified using the Lock Key Writing Sequence.

0: Port configuration lock key not active

1: Port configuration lock key active. GPIOx_LCKR register is locked until the next reset.

LOCK key writing sequence:

Write 1

Write 0

Write 1

Read 0

Read 1 (this read is optional but confirms that the lock is active)

Note: During the LOCK Key Writing sequence, the value of LCK[15:0] must not change.

Any error in the lock sequence will abort the lock.

Bits 15:0 **LCKy**: Port x Lock bit y (y= 0 .. 15)

These bits are read write but can only be written when the LCKK bit is 0.

0: Port configuration not locked

1: Port configuration locked.